$$\hat{H}'_{\mathcal{S}} = \hat{H}_{\mathcal{S}} + \frac{1}{2} M \widetilde{\Omega}^2 \hat{X}^2 = \frac{1}{2M} \hat{P}^2 + \frac{1}{2} M \left[\Omega^2 - 2\gamma_0 \Lambda \right] \hat{X}^2. \tag{5.58}$$